MILITARY SPECIFICATION

INSULATOR DISKS, SEMICONDUCTOR DEVICE GENERAL SPECIFICATION FOR

This specification is approved for use by Rome Air Developement Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

* THIS SPECIFICATION INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION (SEE 6.3). USE MIL-M-38527.

1. SCOPE

1.1 Scope. This specification covers the general requirements for semiconductor device insulator disks fabricated from thermal film plastic, and serviceable in the temperature range from -260°C to +400°C. The specific requirements for a particular type of insulator disk are listed in the applicable detail specification.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-46112 - Plastic Sheet and Strip, Polyimide.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Specification sheets. The individual insulator disk requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this basic specification and the applicable specification sheet, the requirements of the specification sheet shall govern.

3.2 Quality conformance inspection. The insulator disks shall be a product that has been inspected and passed quality conformance inspection (see 4.2).

3.3 Component material. Component material shall be plastic polyimide film, type I conforming to MIL-P-46112.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBRD), Griffiss AFB NY 13440 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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3.4 Design and construction. The insulator disk design, construction, and dimensions shall be as specified (see 3.1).

3.4.1 Dimensions. Dimensions are in inches. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

3.4.2 Tolerances. Tolerance on hole diameters shall be ±.010 inch. Tolerances on all other dimensions shall be ±.015 inch.

3.5 Marking. The insulator disks shall be packaged in plastic bags with a tag attached for marking. Unless otherwise specified, (see 6.2), the marking shall be method I in accordance with MIL-STD-1285 except that insulator disks shall not be marked.

3.6 Workmanship. Insulator disks shall be processed in such a manner as to be uniform in quality and shall be free from surface flaws and other defects that will affect life and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Materials inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the plastic polyimide film used in fabricating the insulator disks are in accordance with type I of MIL-P-46112.

4.2.2 Quality conformance inspection. Inspection of product for delivery shall consist of the examinations in 4.4.

4.2.2.1 Inspection lot. An inspection lot shall consist of all insulator disks of the same part number, produced under essentially the same conditions and offered for inspection at one time.

4.2.2.2 Sampling. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level I. The acceptable quality level (AQL) shall be 1.5 percent.

4.3 Inspection of preparation for delivery. Inspection of preparation for delivery will be in accordance with the instructions of the procuring activity (see 6.2).

4.4 Visual and dimensional inspection. Insulator disks shall be examined to verify the design, construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.5, and 3.6).

5. PREPARATION FOR DELIVERY

Preparation for delivery will be in accordance with the instructions of the procuring activity (see 6.2).

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6. NOTES

6.1 Intended use. Insulator disks covered by this specification are intended for use as electrical insulating components. Many additional sizes and shapes of insulator disks not covered by this specification are now in use. Existing designs under which equipment is being and will continue to be fabricated are such as not to permit complete substitution with insulator disks conforming to the requirements of this specification. These requirements may not be suitable in all cases for direct replacement of existing insulator disks.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet, and the complete part number (see 3.1).
(c) Marking of insulator disks if other than that specified (see 3.5).
(d) Preparation for delivery requirements (see 4.3 and section 5).

* 6.3 Substitution data. The insulator disks specified herein are not for use in design after the date of this specification. They are suitable for use in design contracts current to that date and to support existing military equipment. Insulator disks specified in MIL-M-38527/8 are preferred for design and can be used as substitutes for the inactivated insulated disks of the same configuration and dimensions. The .002 inch thick and .005 inch thick insulator disks are preferred as replacements for the .001 inch thick and .003 inch thick insulator disks, respectively.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Air Force - 17

Review activities:
Air Force - 11, 99
DSA - ES

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Air Force - 17

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